

ZebraFET User Guide

Software version 2.2.0 — guideline: OECD Test Guideline 236 (2025 edition)

This guide follows a Fish Embryo Acute Toxicity (FET) test from the first concentration calculation to the finished report. Each section covers one stage of the assay and introduces the parts of the software you need at that point. If you are looking for a particular screen, the stages map one-to-one onto the entries in the **Go** menu.

The software records observations and computes endpoints. It does not replace the guideline: read OECD TG 236 itself for the biological requirements. The full text is available from **Help > OECD TG 236**.

1. Before you start

The first launch opens a short setup wizard. It asks you to accept the license and to choose where ZebraFET keeps its data. Your work lives under that folder:

- `projects/` — one subfolder per project, each holding the project database and its photographs. This is the folder to back up.
- `registry.db` — the index behind the Projects Hub. It is rebuilt from the project folders, so losing it costs you nothing but the recent list.

You can move the data folder later from **File > Change Data Folder**. Existing projects are not moved; they stay where they are until you import them or move them yourself.

The diagnostic log, `zebrafet.log`, is kept separately in the location your operating system reserves for application data, so it survives a change of data folder. **Help > Open Log Folder** reveals it; you never need its path otherwise.

2. Planning the test

Start from the Projects Hub — the screen ZebraFET opens on. Choose **Create New Project** (or **File > New Project**, Ctrl+N).

Identify the experiment

Fill in the project name, the researcher, the test substance and the unit its concentrations are expressed in (mg/L, μ M, and so on). The unit you enter here labels every axis, table and endpoint from this point on.

Set the **number of days** the test runs. OECD TG 236 specifies 96 h, which is four days of observations.

Two collapsible panels, **Substance Details** and **Water Quality & Conditions**, hold the information the report's Materials and Methods section draws on: CAS number, molar mass, purity, dilution water,

hardness, pH, temperature, photoperiod. None of it is required to proceed, and all of it can be added later from **Edit > Project Settings**, but a report is only as complete as the record behind it.

Define the concentration groups

Below the identification fields, set how many groups of each kind the test uses:

Group	Purpose
Negative control	Dilution water only. Required — TG 236 validity is judged against it.
Concentration	The test substance dilution series.
Solvent control	Required whenever a carrier solvent is used.
Positive control	4 mg/L 3,4-dichloroaniline, run at least twice a year.

The concentration plan table grows as you change these counts. Each row carries a group ID, its type, its nominal concentration, the number of wells it needs and a color used to identify it on the plate.

To build a dilution series, enter the highest concentration and a dilution factor, then press **Generate Series**. The table fills from the top concentration downward. TG 236 asks for a geometric series with a separation factor no greater than 2.2; the software does not enforce this, so check the spacing the generator produced.

Set the wells per group — TG 236 requires 20 embryos per test concentration and 20 in the control — and the number of plates. ZebraFET reports the total embryos and the minimum plate count the plan needs, and blocks creation if the plates you asked for cannot hold the plan.

Press **Create Project**. The project folder and its database are created, and the software moves on to the plate layout.

The concentration plan stays editable after creation, from the Concentration Plan screen. Changing it rebuilds the plate layout and the daily scoring sheets, so make structural changes before you start scoring.

3. Laying out the plates

Go > Plate Layout assigns each concentration group to physical wells. The plate grid matches the format chosen at creation — 96-, 48-, 24-, 12- or 6-well.

Working method:

1. Click a group in the list on the left. It becomes the active brush.
2. Click a well to assign it, or drag across the grid to assign a run of wells. Dragging on empty space draws a rectangular selection and assigns everything inside it.
3. **Unassign Well** turns the brush into an eraser.

4. Use the plate tabs to move between plates. **Duplicate Plate** copies the current layout onto the next plate, which is the fastest way to lay out replicate plates.

The counter panel shows assigned wells against planned wells for every group. The layout is complete when each counter reads its target.

Press **Save Layout** to commit. The status bar confirms how many wells were written. Until you save, the layout exists only on screen.

Layout changes are undoable: **Edit > Undo** (Ctrl+Z) reverses a plate clear, duplicate, or assignment.

4. Scoring each day

Go > Experiment View is where the daily observations are recorded. It has a tab per day, and within a day a tab per plate.

Recording a well

Click a well in the grid. The editor on the right shows its current record. Assign one of the five statuses:

Key	Status	Meaning
1	Live Embryo	Alive, not yet hatched.
2	Dead Embryo	Dead before hatching.
3	Live Hatched	Alive and hatched.
4	Dead Hatched	Dead after hatching.
5	Absent (use majority)	No embryo present — excluded from the denominator.

Arrow keys move between wells, so a plate can be scored without touching the mouse: press the number, press the arrow, repeat.

Dead Embryo and **Dead Hatched** reveal the four lethal endpoints of TG 236 — coagulation of the embryo, lack of somite formation, non-detachment of the tail and lack of heartbeat. Record every one you observed; the report tabulates them separately.

For any live embryo, record the sublethal observations that apply. The nine morphological endpoints are offered as checkboxes:

- Yolk sac oedema
- Pericardial oedema
- Spinal curvature (scoliosis)
- Tail / fin malformation
- Head / jaw malformation
- Fin malformation or absence
- Pigmentation abnormalities
- Uninflated swim bladder
- Developmental delay

The notes field takes free text for anything the checkboxes do not cover.

Every edit is written to the project database as you make it. There is no unsaved state to lose.

Water quality

The **Water Quality** button on each day records the physicochemical measurements TG 236 asks you to monitor: temperature, dissolved oxygen, pH and conductivity. These appear in the report's monitoring table. Leaving every field blank clears the day rather than storing an empty row.

Closing the day

Review Day summarizes what changed since the previous day — a check on transcription before you commit.

Finalize Day locks the day and carries irreversible states forward: an embryo dead on day 2 is dead on days 3 and 4, and hatching does not reverse. Days must be finalized in order, and the software will not finalize a day whose records contradict the day before it — a well that was dead and is now alive, for instance. Correct the inconsistencies it lists, then finalize.

Reopen Day for Editing unlocks a finalized day. Because later days were filled from this one, reopening a day also reopens every finalized day after it; the confirmation names them.

5. Documenting with photographs

Go > Photo Documentation proposes what is worth photographing on a given day, based on what you scored. Suggestions come in three kinds:

- **Malformation** — wells showing a recorded sublethal endpoint.
- **Representative Status** — a well typical of its concentration group.
- **Significant Status** — a status that appears in a notable share of a group.

Select a suggestion, then a well within it, and use **Attach Photo** to import one or more image files. The files are copied into the project folder, so the project stays self-contained and can be moved or exported without breaking the links. Thumbnails appear in the gallery below; a photograph can be removed from there, which deletes the copied file.

Photographs reach the report through the **Appendix: photographic documentation** section.

6. Analyzing the results

Go > Results and Analysis computes the endpoints. Choose the analysis day — normally the last — and press **Recalculate**. The analysis runs off the interface thread; the controls re-enable when it finishes.

What the software checks first

A red **TEST INVALID** banner appears when an assessed validity criterion of TG 236 fails: control mortality above the acceptable threshold set in project settings (10% by default), or, on the final day, control hatching below 80%. The banner carries the same criteria the report's validity section applies, so what you see on screen is what the report will say. A failed criterion does not stop the analysis — the numbers are still computed — but the test does not meet the guideline.

Fitting options

- **Model** — auto-selection by AICc, or a fixed 4PL, 3PL (bottom or top fixed) or 2PL log-logistic fit.
- **Abbott's correction** — corrects mortality for control response. When it applies, the summary table gains an Abbott-corrected column.
- **Control comparison** — whether the reference is the pooled controls, the negative control alone, or the solvent control alone. The same reference is used for the validity verdict and for the NOEC, so the two cannot disagree.
- **Multiplicity correction** — the adjustment applied to the pairwise comparisons behind NOEC and LOEC.

Changing any of these marks the results stale; press **Recalculate** again.

What you get

The summary table gives, per group: the wells assigned, the embryos actually scored, deaths, mortality, hatching and malformation rates. Both denominators are shown because they differ whenever a well was empty — the scored count is the denominator behind every percentage.

Ten endpoints are reported:

Endpoint	Note
Model	The curve selected or fixed.
LC50	From the fitted curve, with confidence interval.
LC50 (Spearman-Kärber)	Trimmed non-parametric estimate, independent of the fit.
Slope	Hill slope of the fitted curve.
R ²	Fit quality.
NOEC	Highest concentration not differing from control.
LOEC	Lowest concentration differing from control.
Trend (Cochran-Armitage)	Test for a monotone dose-response.
Sublethal LOEC	Lowest concentration with a raised rate of ≥ 1 abnormality.
Teratogenic Index	LC50/EC50 — separation of lethal from sublethal effect.

Six figures are available as tabs: Mortality Dose-Response, LC50 Time-Series, Mortality Time-Course, Fate Composition, Hatching Rate and Malformation Profile. **LC50 Time-Series** fits the LC50 at every daily timepoint and is computed on demand from **Analysis > LC50 Time-Series**.

Any group that cannot be evaluated — no scored embryos — is named below the table rather than silently dropped.

7. Reporting and export

Everything below is on the **File > Export** menu, and the figures also have buttons above the plot area.

- **Word Report** — the formatted report: methods, validity assessment, results tables, figures and appendices. A picker lets you choose which optional sections to include; the test validity assessment and the lethal endpoint results are always present. Your selection is remembered for the next report.
- **Analysis Tables (CSV)** — a folder of the computed results: per-group summary, endpoints, sublethal tests and effect sizes, ready for a statistics package.
- **Raw Data (CSV)** — one row per well observation. This is the input to the analysis, not its output, and is the right export for reanalysis elsewhere.
- **Current Figure** — the visible figure as PNG, SVG or PDF at 600 dpi. **Edit > Copy Figure** puts the same image on the clipboard.
- **Project Archive (.zfet)** — the whole project, photographs included, as one file for sharing or archiving. **File > Import** opens one on another machine.

8. Keyboard reference

ZebraFET uses each platform's standard sequences, so a few differ between Windows, Linux and macOS. The menus always show the sequence in force on the machine you are running; where the table below gives two, the second is macOS. On macOS, read Ctrl as Command throughout.

Shortcut	Action
Ctrl+N	New project
Ctrl+O	Open project
Ctrl+W	Close project, return to the Hub
Ctrl+S	Flush the database to disk
Ctrl+Z	Undo
Ctrl+Y / Cmd+Shift+Z	Redo
Ctrl+1 ... Ctrl+6	Jump to a stage: Hub, Concentration Plan, Plate Layout, Experiment View, Photo Documentation, Results
F5 / Cmd+R	Recalculate the analysis
F1 / Cmd+?	This guide
Cmd+,	Project settings (macOS; use the Edit menu elsewhere)
1 ... 5	Set the selected well's status
Arrow keys	Move between wells

About Ctrl+S. Edits are committed as you make them, so there is no unsaved work. What Ctrl+S does is flush the database's write-ahead log, leaving the .db file on disk fully up to date. Use it before copying or backing up a project folder. If the flush fails, the software says so — it will not report a save it did not perform.

9. Where things are

Item	Location
Projects	The data folder chosen in setup, under <code>projects/</code>
Log file	Help > Open Log Folder (<code>zebrafet.log</code>)
Data folder	Shown in the setup summary; change from File > Change Data Folder
This guide	Help > User Guide , also as a PDF beside it
OECD TG 236	Help > OECD TG 236
Licenses	Help > Licenses

10. Citing ZebraFET

ZebraFET is released under the GNU General Public License v3.0. Each release is archived with a DOI; the **About** tab carries the DOI and the repository address for the version you are running.